Progress Report

Main goal/purpose for this period's work

The main goal of this period was to get introduced to all the components, software and concepts we have to be familiar with to complete this bachelor project. This includes working to complete the boat in a mechanical sense, electronics, getting familiar with different control system options, simulator options, and all other components.

Planned activities this period

Its planned to complete the mechanical construction of the USV, including mounting the azimuth brackets in the corner of the fiberglass hulls, building the aluminium construction of the boat, building the azimuth rotor assembly, painting the inside of the USV, build the mounting system for the decks, and other small assembly steps.

Its planned to install electronics into the boat, installing whats needed for atleast a first seaside test, but hopefully time for designing most of the electronics.

The planned activities in the period was to finish the preliminary project, mechanical construction of the USV, setting up the stonefish simulator with ardupilot.

Actually conducted activities this period

The mechanical construction of the USV got completed, a set of test electronics mounted on it, and a seaside test was completed. The USV was found to perform in a satisfactory way mechanically, and is now ready for the real electronics to be mounted. The azimuth thrusters are able to rotate a complete rotation in under 0.5 seconds, and are plenty strong enough to fight the thruster. The thrusters are more than strong enough for the USV, as at full power its angled at an unreasonably high upwards pitch.

The Ardupilot software was researched a lot and firmware for the CubeOrangePlus was compiled and uploaded. The control unit was then connected and setup with the GPS with RTK and tested.

MAVROS along with MAVlink router was configured and tested slightly. This essentially connected the ardupilot mavlink system to a ROS2 system.

The simulator (Stonefish Simulator) was started on and is almost finished, the remaining parts involve integrating the Autodrone challenges along with adding common objects. The simulator has been successfully integrated with the Ardupilot autopilot (Software-In-The-Loop). Simulator data is gathered from the stonefish simulator by subscribing to the Odometry, GPS and IMU ROS2 topics. The data is then packed in JSON format and sent over UDP to the ArduPilot SITL python script which simulates the Ardupilot autpilot behavior, SITL then responds with a list of control inputs for the USV which is then published to the motors and servos.

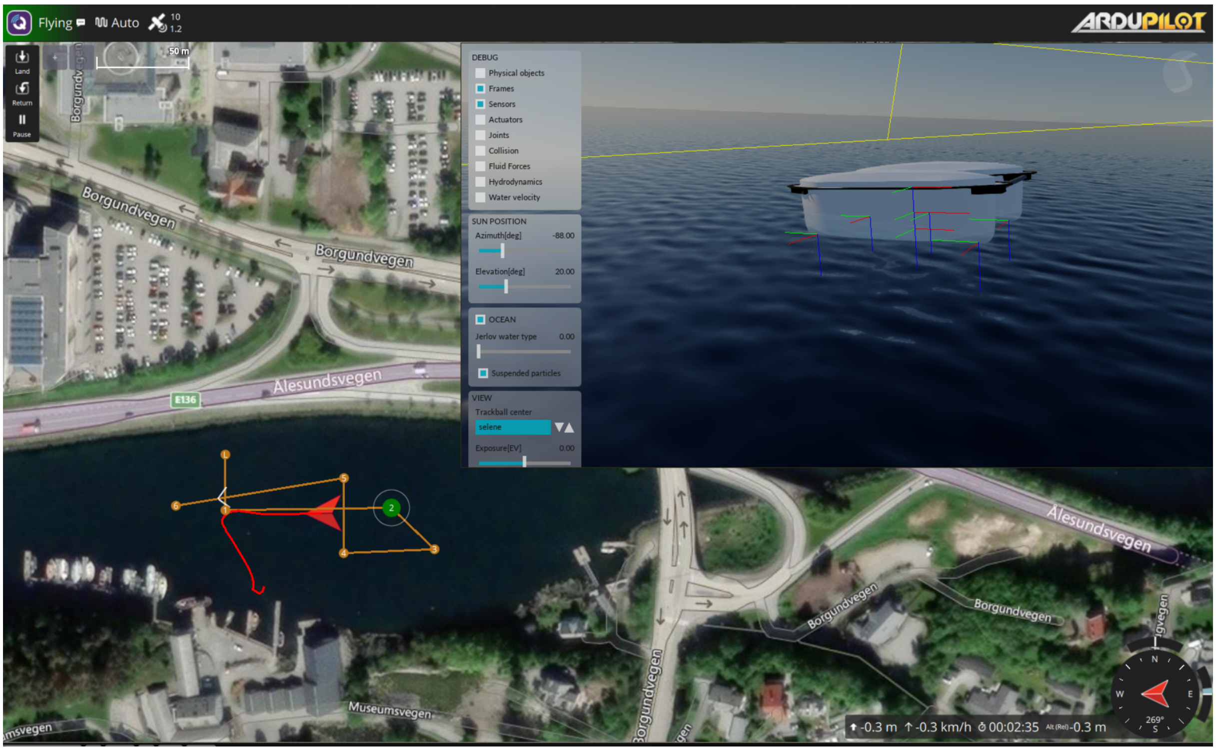

Figure 1: SITL

Additionally a simple Lua script was developed for controlling the simulated USV, however there is a heading mismatch between the simulator and the ArduPilot- or QGroundControl software that causes the displayed heading to be rotated 180 degrees, as seen on figure 1 (The USV is travelling towards the point marked in green).

QGroundControl was built as a custom project, the result was not that much different from the standard QGC software, but it still needs alot of research such that we can be able to place markers purely from software.

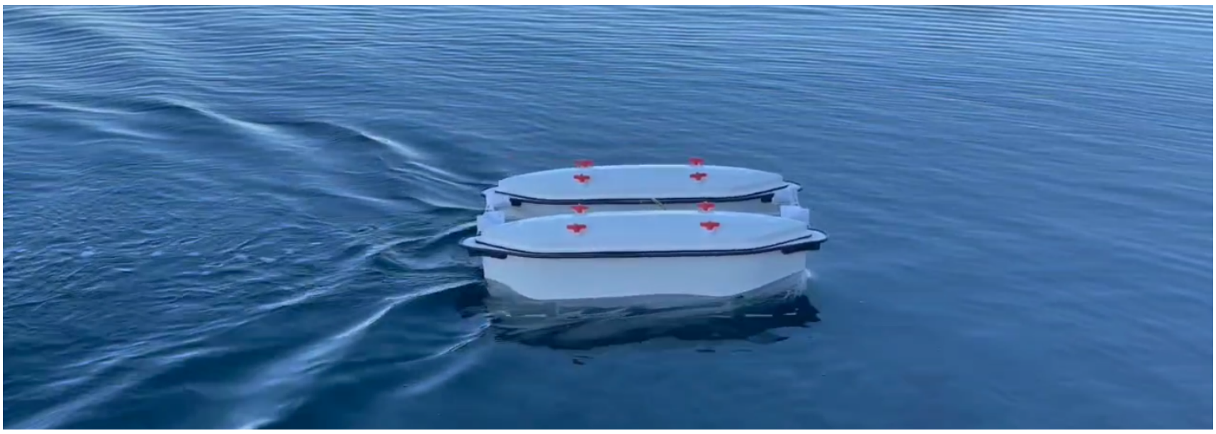

Figure 2: First Seaside Test of Seadrone

Description of/justification for potential deviation between planned and real activities

The electronics didn't get completed to the extent we wanted, this was mainly due to the lack of time to complete, and also the change of control systems through this period.

Description of/justification for changes desired in the project's content or further plan

The plan was originally to design a fully custom system without relying on the Ardupilot and QGroundControl frameworks, a NMEA parser for the GPS was built from scratch and a custom GUI was prototyped with a map and client. But after the first supervision meeting we got a reality check and realized that the amount of work required was unobtainable and would've probably resulted in a unfinished product. The framework Ardupilot was then adapted for the project along with the QGroundControl software.

Main experience from this period

The main experience we got from this project period is that things doesn't always go to plan, and we have to plan contingencies to counter this. We might also have a slight problem with time optimism. Therefore, looking for ways to complete the goals of the project in simpler and more focused ways in later stages might be good to plan for.

Main purpose/focus next period

The main goals for the next period is to continue working on the simulator, integrating the CubeOrange onto the USV, and finishing the electronics. Along with these goals, we also plan to have a test day.

Planned activities next period

- Finish simulator, add buoys and start working on implementing the Autodrone challenges.
- Complete the electronics in a way where its possible to run ArduPilot with navigation.
- Work more on interfacing with the USV such that it is possible to send waypoints and modes from ROS2.

Other / Wish for counseling

Supervision meeting summary

Combined Total Hours Worked This Period (Includes hours before first supervision meeting): **311 HOURS**

Ørjan T. Lynnes

Ørjan Tennøy Lynnes

Liam Følsvik

Liam Harris Følsvik

Date: **26 January 2026**